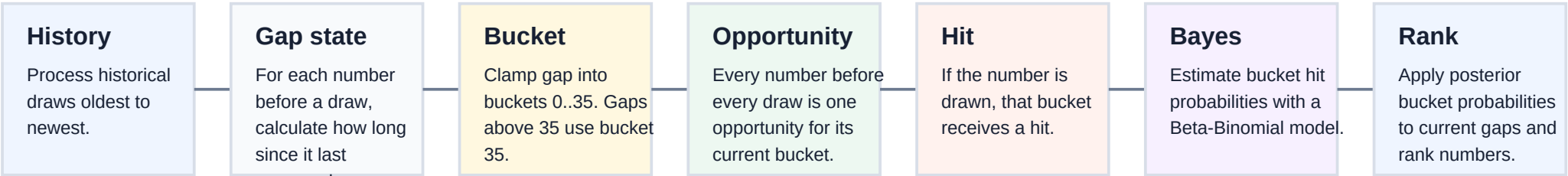


PyLotto Bayesian Markov Calculation

Beta-Binomial gap-state model for ranking next-draw numbers



Model constants from the export script

MAX_GAP_BUCKET 35	NUMBERS_PER_DRAW 6	NUMBER_COUNT 49	PRIOR_STRENGTH 8.0
POSTERIOR_DRAWS 1000	TUNE_DRAWS 1000	CHAINS 2	RANDOM_SEED 20260619

Gap formula

Before a draw
gap = drawIndex if never seen, otherwise drawIndex - 1 - previousSeenIndex

Bayesian Posterior

How opportunities and hits become bucket probabilities

Training counts

The first draw initializes lastSeen and is not counted as training. For every later draw, every number 1..49 contributes one opportunity to its current gap bucket. Drawn numbers also contribute one hit.

Prior

Base probability

$\text{baseProbability} = \text{numbersPerDraw} / \text{numberCount} = 6 / 49$

Beta prior

$\alpha = \text{priorStrength} * \text{baseProbability}$; $\beta = \text{priorStrength} * (1 - \text{baseProbability})$

Likelihood and posterior sampling

Bucket probability

$p_{\text{bucket}} \sim \text{Beta}(\alpha, \beta)$, one probability for each gap bucket 0..35.

Observed hits

$\text{hits_bucket} \sim \text{Binomial}(\text{opportunities_bucket}, p_{\text{bucket}})$

Posterior summary per bucket

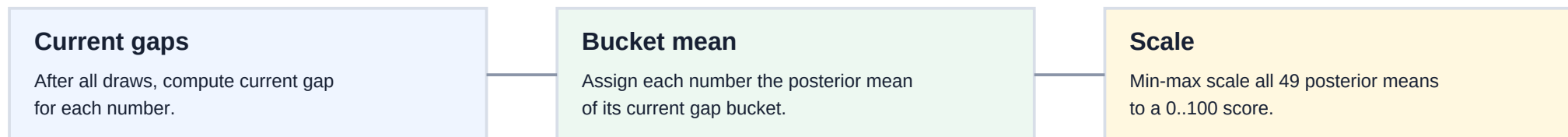
The export stores posterior mean, median, 5th percentile, and 95th percentile. The 5th and 95th percentiles form the displayed 90% credible interval.

The generated JSON lives at <web/src/data/bayesian-markov-score.json>.

Bayesian Ranking and Scores

How posterior bucket means become next-draw rankings

Current prediction



Score formula

Scaled score

$$\text{score}(n) = (\text{posteriorMean}(n) - \text{minMean}) / (\text{maxMean} - \text{minMean}) * 100$$

Band labels



Sort order

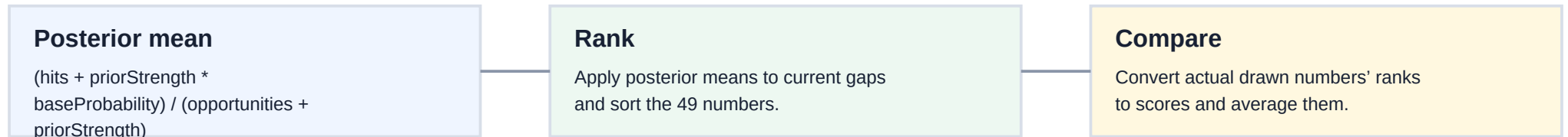
Numbers are ranked by score descending, posterior mean descending, current gap descending, then smaller number. The top six become the Bayesian top numbers.

Bayesian Backtesting

How Bayesian scores are evaluated in Draw Scores and Score Graphs

Previous-history-only scoring

For each draw after the first, the app rebuilds simple posterior means from previous opportunities and hits using the same prior. It ranks numbers using only earlier draws, then scores the actual draw.



Backtest metrics



The UI report uses the generated PyMC JSON for the current model display, while backtests use the closed-form posterior mean for each historical prefix.